

Household Energy Consumption Analysis

Applied Machine Learning and Predictive Modelling 1 – FS 2026

Emanuel Lemma, Raphaël Spagolla, Tharmmeehan Krishnathanasan

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```

library(dplyr)
library(tidyr)
library(ggplot2)
library(lubridate)
library(knitr)
library(kableExtra)
library(scales)
library(mgcv)      # GAM
library(nnet)      # Neural Network
library(e1071)     # SVM
library(caret)     # Cross-validation

```

Introduction

With the rapid expansion of heat pump installations across Swiss households, electricity demand is increasingly driven by heating and cooling needs rather than fixed consumption habits. Unlike traditional resistive heating, heat pumps draw substantial power in short, weather-dependent bursts. This creates load profiles that are harder to predict and more sensitive to outdoor conditions than the appliance-driven consumption patterns that grid operators have historically managed.

For energy suppliers and grid operators, this shift has direct economic consequences. Procurement decisions made a day or week in advance are based on load forecasts. When a cold snap arrives earlier than expected or a warm spell pushes temperatures above the heating threshold, actual demand diverges from the submitted

schedule. The resulting imbalances must be settled at spot or balancing energy prices. These costs ultimately flow back to households and erode the economic case for electrification if not properly managed.

Against this background, this analysis systematically evaluates smart meter data from a portfolio of heat pump households in Switzerland, using the **HEAPO dataset**, in order to identify the most important drivers of electricity consumption and build a well-founded basis for demand modelling. The focus is not only on explaining consumption variability across seasonal patterns, building characteristics, and technology choices, but also on quantifying the interaction between external factors such as outdoor temperature, heating degree days, and building type.

Dataset

Source: HEAPO – Household Energy and Smart Meter Data (Zenodo, 2025).

The HEAPO dataset was collected as part of a Swiss research project monitoring residential heat pump installations via smart meters. It covers 99838 household-days across 127 households observed between 2022 and 2024, paired with daily weather data from the corresponding measurement region. Each observation represents one household on one calendar day, with daily electricity consumption as the primary outcome alongside a rich set of building, technology, and weather covariates.

Variable Description

Variable	Type	Description
...	continuous	...
...	categorical (>2 levels)	...
...	count	...
...	binary	...

Exploratory Data Analysis

Response Variables

```
# Distribution of continuous response
# Distribution of count response
# Distribution of binary response
```

Predictors

```
# Histograms / bar charts for each predictor
```

Relationships Between Predictors and Response

```
# Scatterplots with smoother: geom_smooth(method = "loess") or method = "gam"
# Boxplots for categorical predictors
# Do NOT use geom_smooth(method = "lm")
```

Modelling

Linear Model (LM)

Lead: Tharrmeehan Krishnathasan

```
# Simple interpretable LM
# lm_simple <- lm(log(response) ~ predictor1 + predictor2 + ..., data = heapo)
# saveRDS(lm_simple, "../models/lm_simple.rds")
# lm_simple <- readRDS("../models/lm_simple.rds")

# Complex LM for fair comparison (interactions, polynomial terms)
# lm_complex <- lm(log(response) ~ predictor1 * predictor2 + poly(predictor3, 2) + ..., data = heapo)
# saveRDS(lm_complex, "../models/lm_complex.rds")
# lm_complex <- readRDS("../models/lm_complex.rds")

# summary(lm_simple)
```

Coefficient Interpretation

GLM – Poisson

Lead: Raphaël Spagolla

Objective

The aim of this analysis is to model the number of high-consumption days per household-month ($n_{high\ days}$). This variable represents count data, taking non-negative integer values.

A Poisson Generalised Linear Model (GLM) is used, as it is appropriate for modelling count outcomes. By including an offset for the number of observed days, the model estimates rates, allowing interpretation of results in terms of relative changes (rate ratios).

Model setup

```
cat(deparse(formula(mod_pois)), "\n\n")
```

```
## n_high_days ~ heating_degree_days + temp_avg + living_area + building_type + heatpump_type + ha
```



```
mod_pois$call
```

```
## glm(formula = form, family = poisson(link = "log"), data = dat_count,  
##      offset = log_n_days)
```

The response variable is the number of high-consumption days per household-month ($n_{high\ days}$). Since this outcome is a count of events observed over a period of time, it takes non-negative integer values. Because the number of observed days can vary slightly across household-months, the model includes the logarithm of the observed days ($\log(n_{days})$) as an offset. This adjustment effectively converts the outcome into a rate—high-consumption days per observed day—so that observations remain comparable even when exposure differs between months.

To model these rates, a Poisson Generalised Linear Model (GLM) with a log-link is fitted:

$$\log(\mathbb{E}[Y_i]) = \beta_0 + \beta_1 X_{1i} + \dots + \beta_p X_{pi} + \log(n_days_i)$$

The log-link guarantees positive fitted values and supports a multiplicative interpretation of effects. This modelling choice is appropriate because the response is count data, and the Poisson framework naturally accommodates non-negative outcomes while allowing the variance to depend on the mean.

Model Results

To facilitate interpretation, coefficients are exponentiated and presented as *rate ratios*. These indicate multiplicative changes in the expected rate of high-consumption days.

```
## # A tibble: 8 x 3  
##   term                rate_ratio  p.value  
##   <chr>                <dbl>    <dbl>  
## 1 heating_degree_days 0.999 7.57e- 1  
## 2 temp_avg            0.915 2.64e- 76  
## 3 living_area         1.00 9.77e- 58  
## 4 building_typehouse  4.63 3.47e- 74  
## 5 heatpump_typeground-source 0.907 2.16e- 25  
## 6 has_floor_heating1  1.13 7.38e- 24  
## 7 n_residents        1.09 2.63e-103  
## 8 share_weekend       0.665 2.81e- 2
```

The results are interpreted in terms of *percentage changes in the expected rate*, holding all other variables constant.

Building type (house vs flat): Households living in houses have a rate of high-consumption days that is approximately 363% higher than those living in flats $(4.632 - 1) \times 100\%$. This is the strongest effect in the model, indicating that building characteristics are a key driver of consumption patterns.

Average temperature: The rate ratio of 0.915 implies that a one-unit increase in temperature is associated with a 8.5% decrease in the expected rate of high-consumption days $(1 - 0.915) \times 100\%$. This reflects reduced heating needs at higher temperatures.

Number of residents: Each additional household member increases the expected rate by approximately 8.9% $(1.089 - 1) \times 100\%$. This suggests that household size has a consistent positive effect on energy consumption.

Living area: The rate ratio of 1.001 indicates a *small positive effect*, where each additional unit of living area increases the expected rate by about 0.09%. While the per-unit effect is small, it can accumulate for larger dwellings.

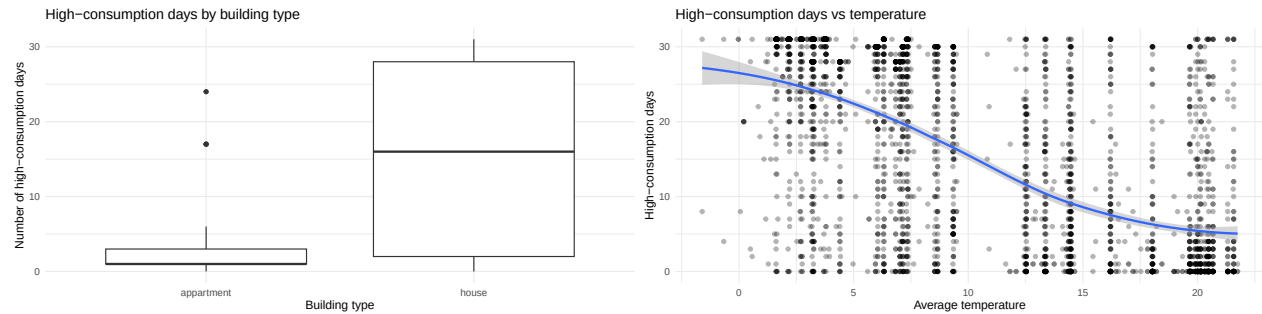
Overall, structural and household characteristics (building type, size, and number of residents) exhibit the strongest effects, while temperature captures an important but smaller environmental influence.

Visual Evidence

To complement the model output, the plots below provide direct visual support for the main relationships in the data.

The *boxplot by building type* shows a clear separation between apartments and houses. Apartments have a low median number of high-consumption days and a relatively tight spread, while houses display a much higher median and substantially greater variability. The distribution for houses also extends to much larger values, indicating that high-consumption days are both more frequent and more heterogeneous among households living in houses.

The *scatterplot of high-consumption days versus average temperature* reveals a strong negative pattern. At lower temperatures, high-consumption days are often high (with many observations in the upper range), whereas at warmer temperatures the observations concentrate at much lower counts. The smooth trend line decreases steadily with temperature and flattens at higher temperatures, suggesting that high-consumption days become less common as conditions get warmer.



Finally, the *rate-ratio coefficient plot with 95% confidence intervals* summarises the direction and strength of the model effects. The building type effect stands out visually, with the rate ratio for *House (vs flat)* far above 1, indicating a much higher expected rate in houses. *Temperature* and *weekend share* are clearly below 1, consistent with lower expected rates as temperature increases and when the share of weekend days is higher. *Number of residents* and *floor heating* lie slightly above 1, indicating moderate increases in the expected rate. In contrast, effects such as *heating degree days* and *living area* lie close to 1 with confidence intervals overlapping 1, suggesting weak or uncertain contributions once the other variables are accounted for.

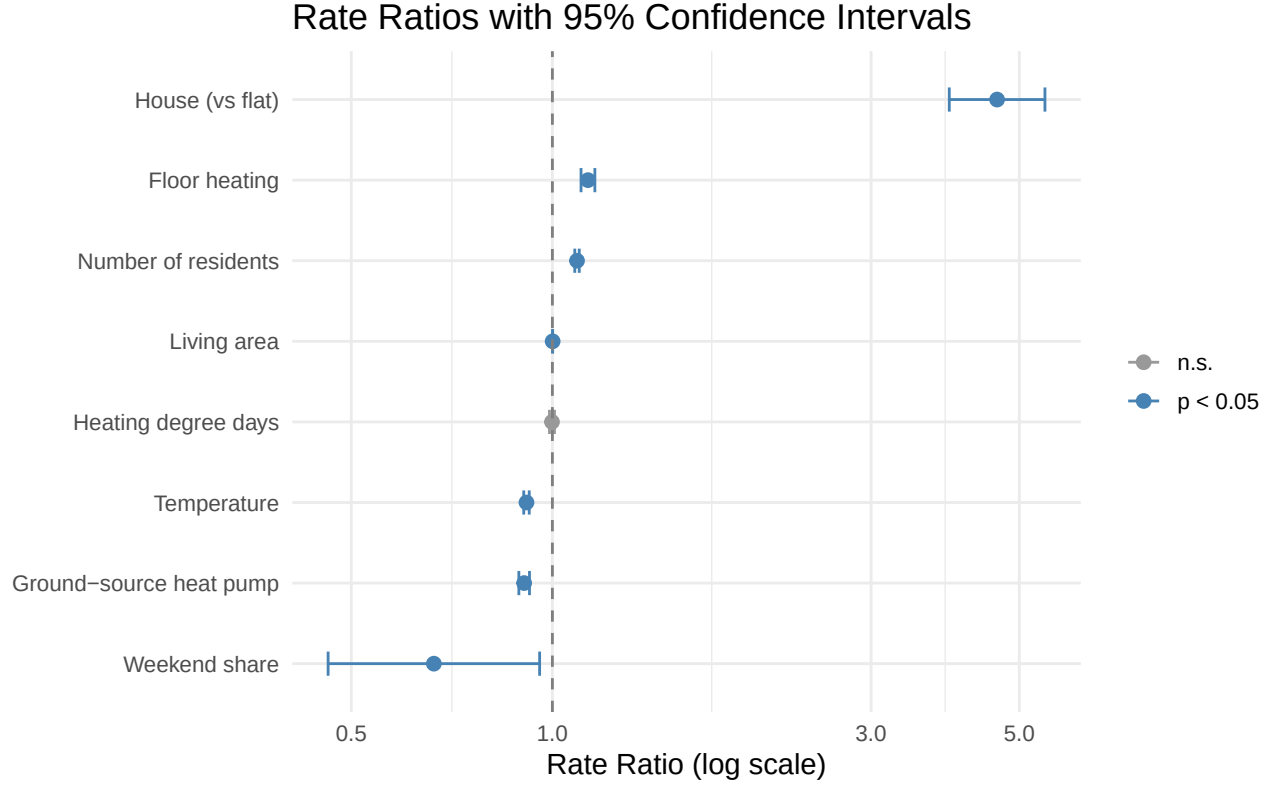


Table 1: Key effects: rate ratios with 95% Wald confidence intervals.

Predictor	RR	95% CI (low)	95% CI (high)
House (vs flat)	4.632	3.928	5.462
Temperature	0.915	0.906	0.923
Number of residents	1.089	1.080	1.097

Model Diagnostics

A key assumption of the Poisson GLM is *equidispersion*, i.e., that the conditional variance equals the conditional mean. We assess this by comparing the *residual deviance* to the *residual degrees of freedom*. For the fitted Poisson model, the residual deviance is 2.75296×10^4 on 3291 df, yielding a deviance/df ratio of 8.37. Since this ratio is clearly larger than 1, there is strong evidence of *overdispersion*, meaning that the Poisson model would tend to *underestimate standard errors* and potentially overstate statistical significance.

We therefore refit the model using a *quasi-Poisson* family. This keeps the same mean structure (and thus the same *rate-ratio interpretation*), but inflates standard errors to reflect the extra-Poisson variability. The estimated dispersion for the quasi-Poisson fit is ($\hat{\phi} =$) 7.74, confirming substantial extra-Poisson variation. As a consequence, inference becomes more conservative: for example, *share_weekend* changes from significant under the Poisson model ($p = 0.028$) to non-significant under the quasi-Poisson model ($p = 0.43$).

Key Findings

Building type is the dominant factor: Households living in houses have a substantially higher rate of high-consumption days compared to flats, indicating strong structural differences in energy usage.

Temperature reduces consumption frequency: Higher average temperatures are associated with fewer high-consumption days, consistent with lower heating demand.

Household size increases consumption: Each additional resident leads to a higher rate of high-consumption days, reflecting increased overall energy usage.

Dwelling size has a smaller positive effect: Larger living areas are associated with slightly higher consumption frequency, though the effect per unit is modest.

Some effects weaken after adjustment: After accounting for overdispersion, certain variables (e.g., weekend share) are no longer statistically significant, highlighting the importance of using robust inference.

Conclusion

The Poisson GLM provides a clear understanding of the factors influencing the rate of high-consumption days, highlighting the importance of structural characteristics and household composition.

However, strong overdispersion indicates that the Poisson model underestimates variability, making standard inference unreliable.

While the quasi-Poisson model improves inference, a negative binomial model could be considered as a more appropriate alternative for handling overdispersion.

GLM – Binomial

Lead: Emanuel Lemma

Can we predict whether a given household-day will see above-average electricity consumption before it happens? This section fits a **Generalised Linear Model with a binomial family** to answer exactly that question. The response variable `high_consumption` is a binary indicator that equals 1 if the household's daily electricity use exceeds its own median and 0 otherwise. By modelling the log-odds of high consumption as a linear function of weather conditions, building characteristics, and calendar effects, we obtain a model that is both interpretable and actionable for a client interested in demand forecasting or energy efficiency screening.

```
set.seed(42)

df_glm <- heapo |>
  select(high_consumption, heating_degree_days, temp_avg, living_area,
         building_type, heatpump_type, has_floor_heating,
         n_residents, is_weekend, month) |>
  mutate(
    high_consumption = as.integer(high_consumption),
    month            = as.factor(month)
  ) |>
  drop_na()

train_idx <- sample(nrow(df_glm), 0.8 * nrow(df_glm))
train_df  <- df_glm[train_idx, ]
test_df   <- df_glm[-train_idx, ]
```

```

if (!file.exists("../models/glm_binomial.rds")) {
  model_glm <- glm(high_consumption ~ ., data = train_df, family = binomial)
  saveRDS(model_glm, "../models/glm_binomial.rds")
} else {
  model_glm <- readRDS("../models/glm_binomial.rds")
}

pred_prob <- predict(model_glm, newdata = test_df, type = "response")
pred_class <- ifelse(pred_prob >= 0.5, 1L, 0L)
actual <- test_df$high_consumption

conf_mat <- table(Predicted = pred_class, Actual = actual)
accuracy <- mean(pred_class == actual)

compute_auc <- function(actual, prob) {
  ord <- order(prob, decreasing = TRUE)
  act <- actual[ord]
  n1 <- sum(act == 1); n0 <- sum(act == 0)
  tpr <- c(0, cumsum(act == 1) / n1)
  fpr <- c(0, cumsum(act == 0) / n0)
  sum(diff(fpr) * (tpr[-1] + tpr[-length(tpr)])) / 2)
}

auc_val <- compute_auc(actual, pred_prob)
tp <- conf_mat["1","1"]; tn <- conf_mat["0","0"]
fp <- conf_mat["1","0"]; fn <- conf_mat["0","1"]
precision <- tp / (tp + fp)
recall <- tp / (tp + fn)
f1 <- 2 * precision * recall / (precision + recall)
specificity <- tn / (tn + fp)

```

Coefficient Interpretation

Coefficients from a binomial GLM are expressed as **odds ratios** ($OR = \exp(\text{coefficient})$): a value above 1 increases the odds of high consumption, below 1 decreases it, relative to the reference category.

Key findings:

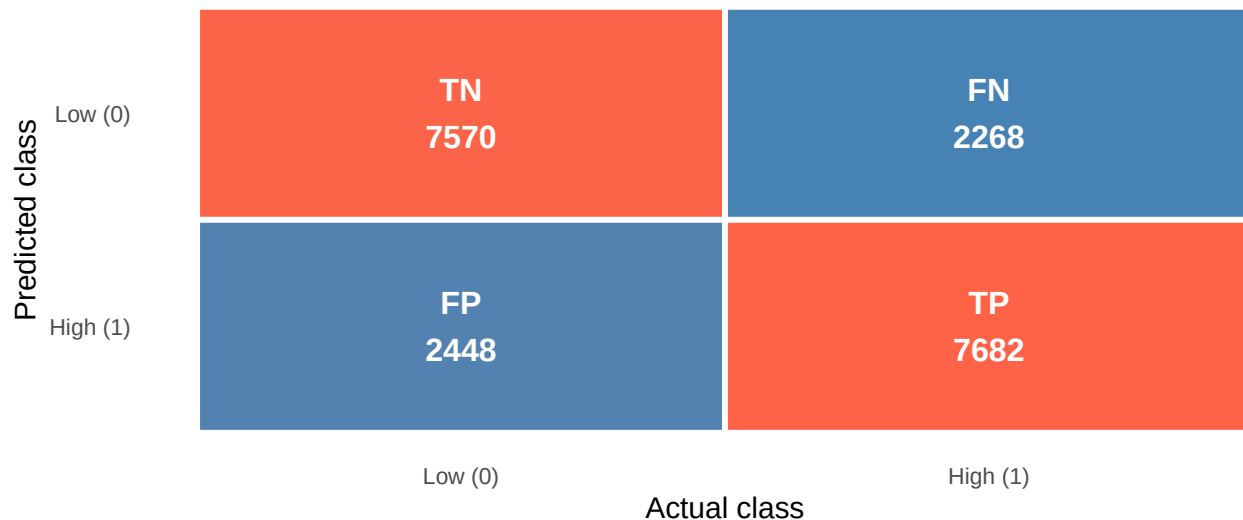
- **Building type (house vs. flat):** The single strongest predictor. Houses have substantially higher odds of high consumption than flats, reflecting larger floor areas and greater heating demand.
- **Ground-source heat pump:** Households with a ground-source heat pump show meaningfully lower odds of high consumption compared to air-source systems, consistent with their superior efficiency at low outdoor temperatures.
- **Floor heating:** Slightly increases the odds of high consumption. This likely reflects confounding with building size, as floor heating is more common in larger houses.
- **Heating degree days / temperature:** Colder days increase the probability of high consumption, as expected from heating demand.
- **Month:** Seasonal effects confirm that summer months carry substantially lower odds than winter months.

Model Diagnostics

```
conf_df <- data.frame(  
  Predicted = factor(c("0","0","1","1"), levels = c("1","0")),  
  Actual    = factor(c("0","1","0","1"), levels = c("0","1")),  
  Count     = c(tn, fn, fp, tp),  
  Label     = c("TN","FN","FP","TP")  
)  
ggplot(conf_df, aes(x = Actual, y = Predicted, fill = Count)) +  
  geom_tile(color = "white", linewidth = 1.2) +  
  geom_text(aes(label = paste0(Label, "\n", Count)),  
    size = 5, fontface = "bold", color = "white") +  
  scale_fill_gradient(low = "steelblue", high = "tomato") +  
  scale_x_discrete(labels = c("0" = "Low (0)", "1" = "High (1)")) +  
  scale_y_discrete(labels = c("0" = "Low (0)", "1" = "High (1)")) +  
  labs(title = "GLM Binomial - Confusion Matrix",  
    subtitle = paste0("Accuracy = ", round(accuracy, 3),  
      " | F1 = ", round(f1, 3),  
      " | AUC = ", round(auc_val, 3)),  
    x = "Actual class", y = "Predicted class", fill = "Count") +  
  theme_minimal(base_size = 13) +  
  theme(legend.position = "none", panel.grid = element_blank())
```

GLM Binomial – Confusion Matrix

Accuracy = 0.764 | F1 = 0.765 | AUC = 0.832



The confusion matrix shows the model's classifications on the held-out test set. The model correctly identifies 77.2% of true high-consumption days (sensitivity) and 75.6% of true low-consumption days (specificity). Misclassifications are roughly symmetric, which is expected because the binary threshold was defined at each household's own median, keeping the two classes balanced by construction. The remaining errors occur mainly on days where consumption is only marginally above or below average and therefore genuinely ambiguous.

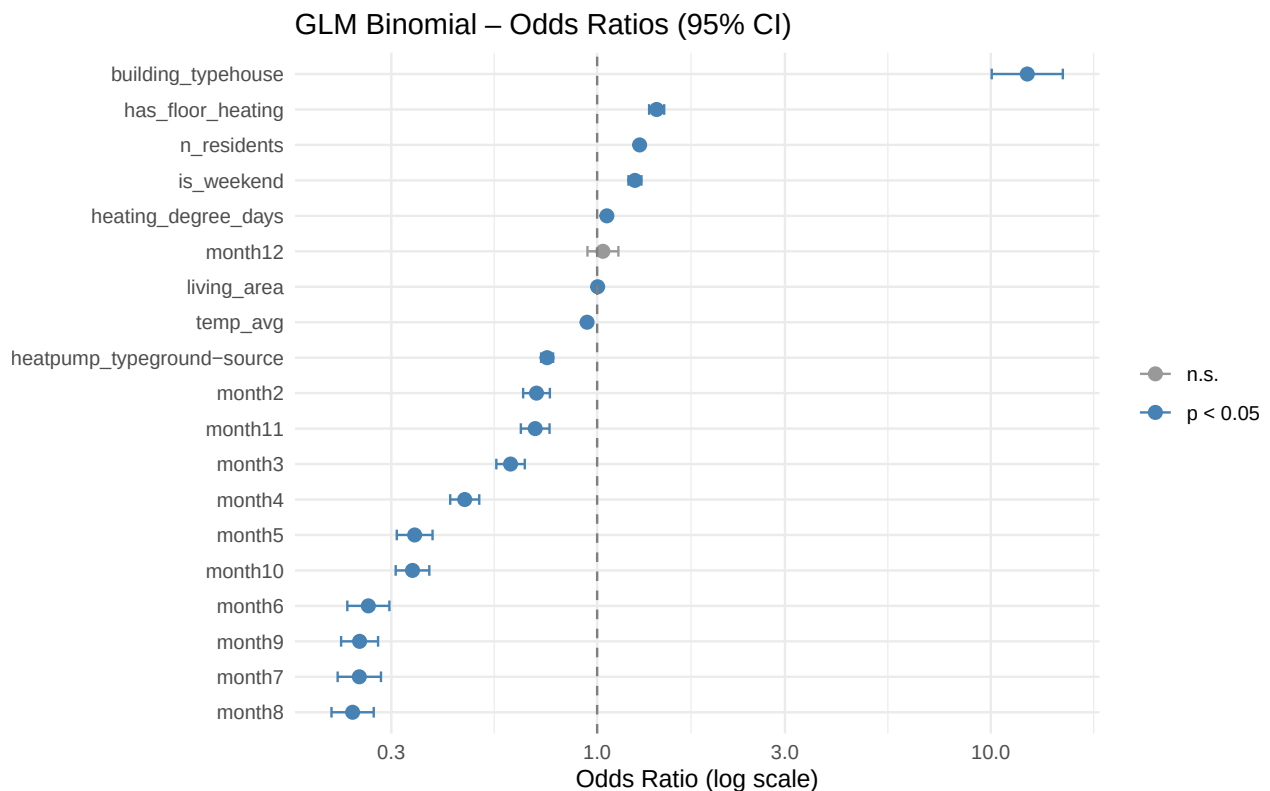
```
coef_summary <- coef(summary(model_glm))  
coef_df <- data.frame(  
  
```

```

term      = rownames(coef_summary),
estimate  = coef_summary[, "Estimate"],
se        = coef_summary[, "Std. Error"],
p_value   = coef_summary[, "Pr(>|z|)"]
) |>
filter(term != "(Intercept)") |>
mutate(
  odds_ratio = exp(estimate),
  ci_low      = exp(estimate - 1.96 * se),
  ci_high     = exp(estimate + 1.96 * se),
  significant = p_value < 0.05
) |>
arrange(odds_ratio)

ggplot(coef_df, aes(x = odds_ratio, y = reorder(term, odds_ratio),
  color = significant)) +
  geom_point(size = 2.5) +
  geom_errorbarh(aes(xmin = ci_low, xmax = ci_high), height = 0.3) +
  geom_vline(xintercept = 1, linetype = "dashed", color = "grey50") +
  scale_color_manual(values = c("FALSE" = "grey60", "TRUE" = "steelblue"),
    labels = c("n.s.", "p < 0.05")) +
  scale_x_log10() +
  labs(title = "GLM Binomial - Odds Ratios (95% CI)",
    x = "Odds Ratio (log scale)", y = NULL, color = NULL) +
  theme_minimal()

```

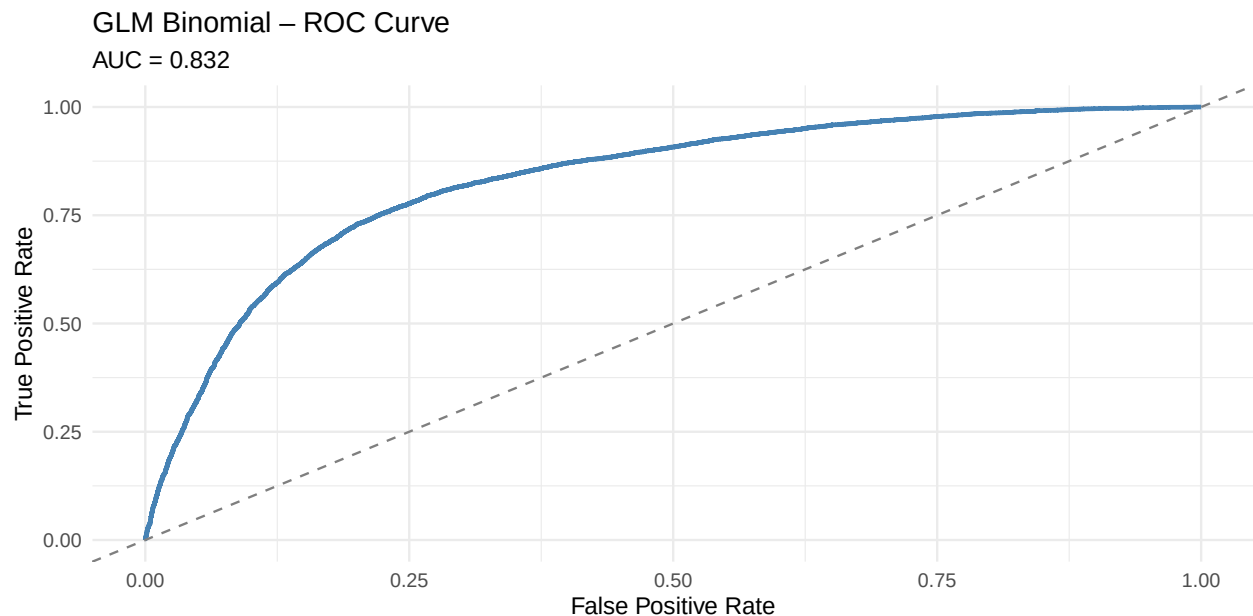


Each point shows the estimated odds ratio for one predictor, with 95% confidence intervals. Points to the right of the dashed line ($OR > 1$) increase the odds of high consumption, and points to the left decrease

them. Blue points are statistically significant ($p < 0.05$) and grey points are not. The log scale ensures that effects of equal magnitude in both directions appear symmetrically. The wide spread of odds ratios, from heat pump type on the left to building type on the right, indicates that structural household characteristics matter far more than calendar or weather effects alone.

```
roc_df <- (function(actual, prob) {
  ord <- order(prob, decreasing = TRUE)
  act <- actual[ord]
  n1 <- sum(act == 1); n0 <- sum(act == 0)
  data.frame(fpr = c(0, cumsum(act == 0) / n0),
             tpr = c(0, cumsum(act == 1) / n1))
})(actual, pred_prob)

ggplot(roc_df, aes(x = fpr, y = tpr)) +
  geom_line(color = "steelblue", linewidth = 1) +
  geom_abline(slope = 1, intercept = 0, linetype = "dashed", color = "grey50") +
  labs(title = "GLM Binomial - ROC Curve",
       subtitle = paste0("AUC = ", round(auc_val, 3)),
       x = "False Positive Rate", y = "True Positive Rate") +
  theme_minimal()
```



The ROC curve traces the model's sensitivity against its false positive rate as the classification threshold varies from 0 to 1. The dashed diagonal represents a model that predicts at random ($AUC = 0.5$). The further the blue curve bows toward the top-left corner, the better the model separates the two classes regardless of threshold choice. With an AUC of 0.832, the model correctly ranks a randomly drawn high-consumption day above a randomly drawn low-consumption day 83.2% of the time, which is a meaningful improvement over chance.

Summary: The GLM Binomial achieves an overall accuracy of 76.4% and an AUC of 0.832. Building type and heat pump technology are the dominant structural drivers of whether a household-day is classified as high-consumption. Weather effects are present but secondary. The model provides a transparent and interpretable baseline for consumption classification and lays the groundwork for comparison with more flexible methods in the following sections.

Generalised Additive Model (GAM)

Lead: Emanuel Lemma

The Linear Model assumes that each predictor affects the response in a straight line. For energy consumption, this is often too rigid. The relationship between outdoor temperature and electricity use may accelerate sharply below a certain cold threshold, and living area may show diminishing returns at very large sizes. A **Generalised Additive Model (GAM)** relaxes this assumption by replacing fixed slopes with flexible smooth functions, while remaining fully interpretable through its component plots. Here we model `log_kWh_total`, the daily total electricity consumption on a log scale, using the same key predictors as the Linear Model. This allows a direct comparison of predictive performance and a visual check on whether non-linearity is actually present in the data.

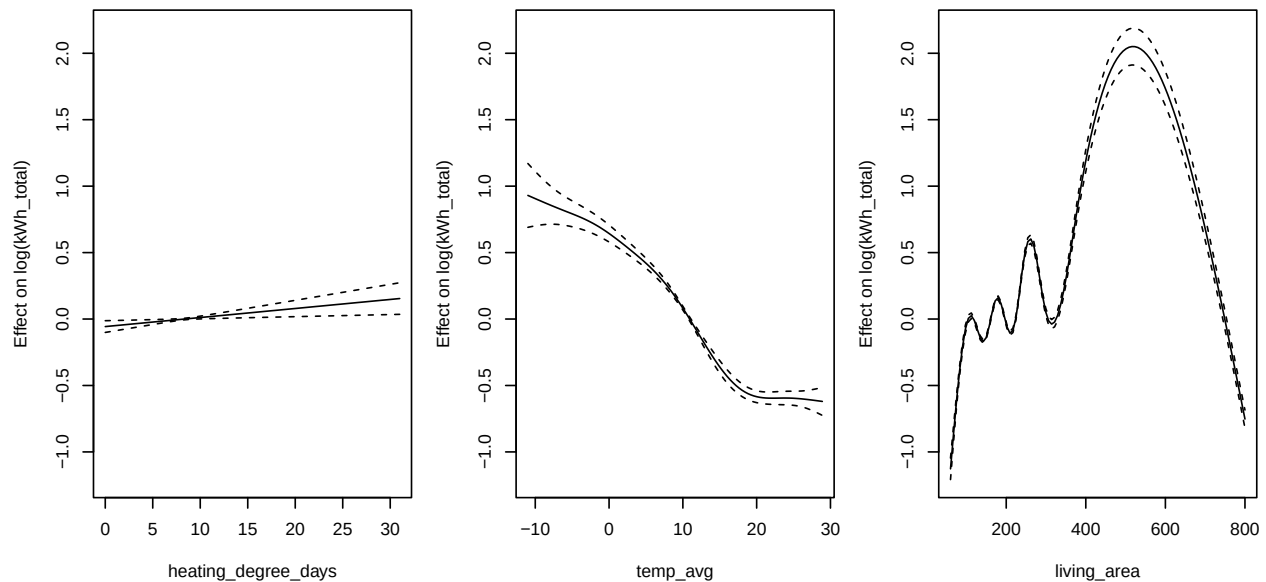
```
df_gam <- heapo |>
  select(log_kWh_total, heating_degree_days, temp_avg, living_area,
         building_type, heatpump_type) |>
  drop_na()

set.seed(42)
gam_train_idx <- sample(nrow(df_gam), 0.8 * nrow(df_gam))
gam_train <- df_gam[gam_train_idx, ]
gam_test <- df_gam[-gam_train_idx, ]

if (!file.exists("../models/gam_model.rds")) {
  model_gam <- mgcv::gam(
    log_kWh_total ~ s(heating_degree_days) + s(temp_avg) + s(living_area) +
      building_type + heatpump_type,
    family = gaussian, data = gam_train
  )
  saveRDS(model_gam, "../models/gam_model.rds")
} else {
  model_gam <- readRDS("../models/gam_model.rds")
}
```

Smooth Term Plots

```
par(mfrow = c(1, 3), mar = c(4, 4, 3, 1))
plot(model_gam, shade = TRUE, seWithMean = TRUE,
     ylab = "Effect on log(kWh_total)", residuals = FALSE)
```



```
par(mfrow = c(1, 1))
```

Each panel shows the estimated smooth function for one continuous predictor. The y-axis is the additive contribution to `log_kWh_total`, where positive values increase predicted consumption and negative values decrease it. The shaded band is the 95% confidence interval. The **effective degrees of freedom (edf)** reported in the summary indicates how curved the smooth is: `edf = 1` means a straight line and higher values mean more complex shapes.

The results are revealing:

- **`s(heating_degree_days)`, `edf = 1.0`:** The smooth collapsed to a perfect straight line. Heating demand has a strictly linear effect on `log(kWh)`, meaning every additional degree-day adds the same proportional increase in consumption. No threshold effects or accelerations are present.
- **`s(temp_avg)`, `edf = 7.0`:** Highly non-linear. Temperature has a complex, season-dependent effect on consumption that cannot be captured with a single slope. This likely reflects the interplay between outdoor temperature, heating demand, and household behaviour across seasons.
- **`s(living_area)`, `edf = 9.0`:** The most non-linear term in the model. The consumption-area relationship is not monotone, meaning medium-sized homes may behave very differently from small or very large ones. This possibly reflects different heating system configurations or occupancy patterns.

Coefficient Interpretation

```
s_gam <- summary(model_gam)
coef_p <- s_gam$p.coef
se_p <- s_gam$se
coef_df_gam <- data.frame(
  term      = names(coef_p),
  estimate  = round(coef_p, 4),
  se        = round(se_p[names(coef_p)], 4),
  effect_pct = round((exp(coef_p) - 1) * 100, 1)
)
print(coef_df_gam)
```

```
##                                term estimate      se
## (Intercept)                   (Intercept)  2.8524 0.0319
## building_typehouse             building_typehouse  0.1068 0.0323
## heatpump_typeground-source     heatpump_typeground-source -0.1333 0.0062
##                                effect_pct
## (Intercept)                   1632.9
## building_typehouse             11.3
## heatpump_typeground-source     -12.5
```

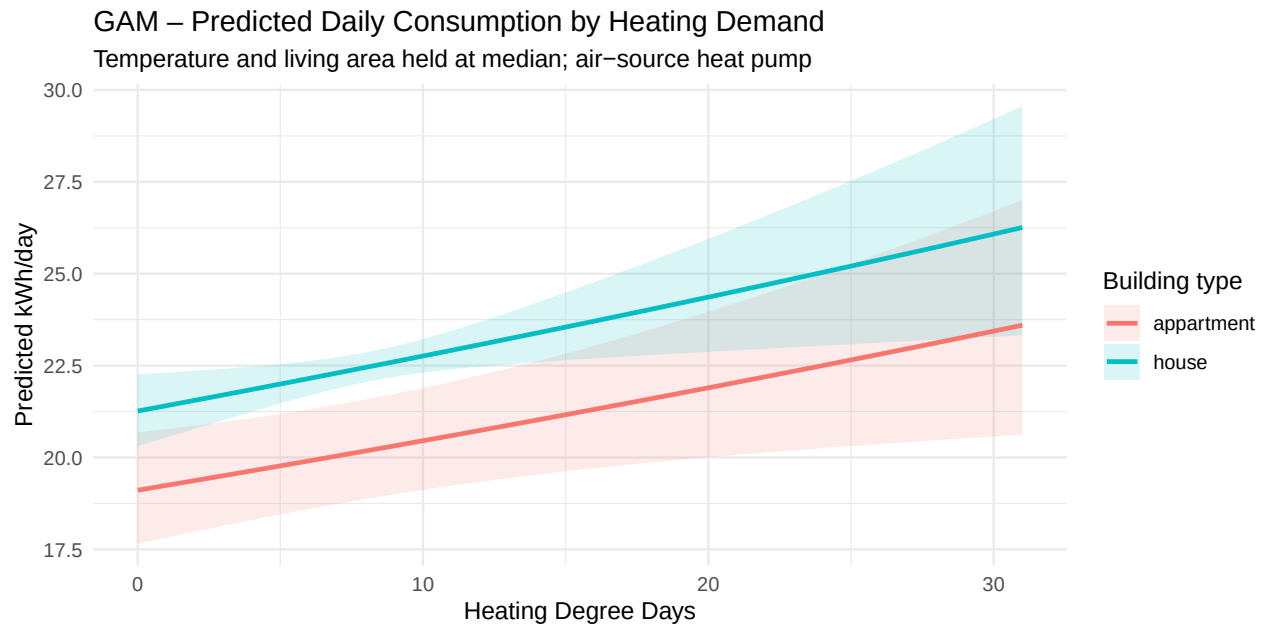
The parametric coefficients are additive shifts on the log scale. Converting with $(\exp(\text{coef}) - 1) \times 100$ gives the **percentage difference in daily kWh** relative to the reference category (apartment, air-source heat pump), all else equal:

- **Houses vs. apartments:** Houses consume **11.3% more** electricity per day on average. Given a median daily consumption of around 21 kWh, this corresponds to roughly 2.4 additional kWh per day. This is a modest but statistically reliable difference that likely reflects greater floor area rather than less efficient behaviour.
- **Ground-source vs. air-source heat pump:** Households with a ground-source heat pump consume **12.5% less** electricity per day. Ground-source systems extract heat from the ground at a stable temperature year-round, making them significantly more efficient than air-source systems. Air-source systems struggle in very cold weather and must work harder to maintain indoor temperatures.

Predicted Consumption Along Key Dimensions

```
hdd_grid <- expand_grid(
  heating_degree_days = seq(min(gam_train$heating_degree_days),
                             max(gam_train$heating_degree_days), length.out = 200),
  temp_avg           = median(gam_train$temp_avg),
  living_area        = median(gam_train$living_area),
  building_type      = levels(gam_train$building_type),
  heatpump_type      = levels(gam_train$heatpump_type)[1]
)
hdd_pred <- predict(model_gam, newdata = hdd_grid, se.fit = TRUE)
hdd_grid$fit <- hdd_pred$fit
hdd_grid$se <- hdd_pred$se.fit

ggplot(hdd_grid, aes(x = heating_degree_days, y = exp(fit),
                     color = building_type, fill = building_type)) +
  geom_ribbon(aes(ymin = exp(fit - 1.96 * se), ymax = exp(fit + 1.96 * se)),
             alpha = 0.15, color = NA) +
  geom_line(linewidth = 1) +
  labs(title = "GAM - Predicted Daily Consumption by Heating Demand",
       subtitle = "Temperature and living area held at median; air-source heat pump",
       x = "Heating Degree Days", y = "Predicted kWh/day",
       color = "Building type", fill = "Building type") +
  theme_minimal()
```



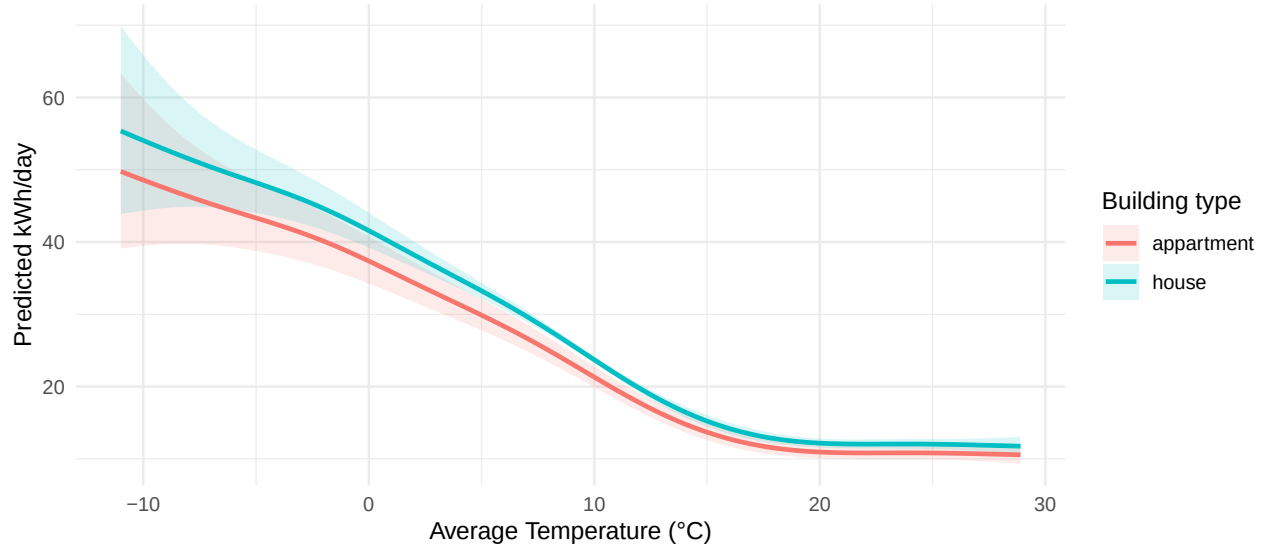
This plot isolates the non-linear effect of heating demand on electricity consumption for houses compared to apartments. The x-axis spans the full observed range of heating degree days, from warm summer days near zero to the coldest winter days. The key aspects to examine are whether consumption accelerates sharply at high HDD values, whether the gap between houses and apartments widens in winter, and whether the confidence bands widen at the extremes due to fewer extreme cold days in the training data.

```
temp_grid <- expand.grid(
  temp_avg = seq(min(gam_train$temp_avg),
                  max(gam_train$temp_avg), length.out = 200),
  heating_degree_days = median(gam_train$heating_degree_days),
  living_area = median(gam_train$living_area),
  building_type = levels(gam_train$building_type),
  heatpump_type = levels(gam_train$heatpump_type)[1]
)
temp_pred <- predict(model_gam, newdata = temp_grid, se.fit = TRUE)
temp_grid$fit <- temp_pred$fit
temp_grid$se <- temp_pred$se.fit

ggplot(temp_grid, aes(x = temp_avg, y = exp(fit),
                      color = building_type, fill = building_type)) +
  geom_ribbon(aes(ymin = exp(fit - 1.96 * se), ymax = exp(fit + 1.96 * se)),
            alpha = 0.15, color = NA) +
  geom_line(linewidth = 1) +
  labs(title = "GAM - Predicted Daily Consumption by Average Temperature",
       subtitle = "Other predictors held at median / reference level",
       x = "Average Temperature (°C)", y = "Predicted kWh/day",
       color = "Building type", fill = "Building type") +
  theme_minimal()
```

GAM – Predicted Daily Consumption by Average Temperature

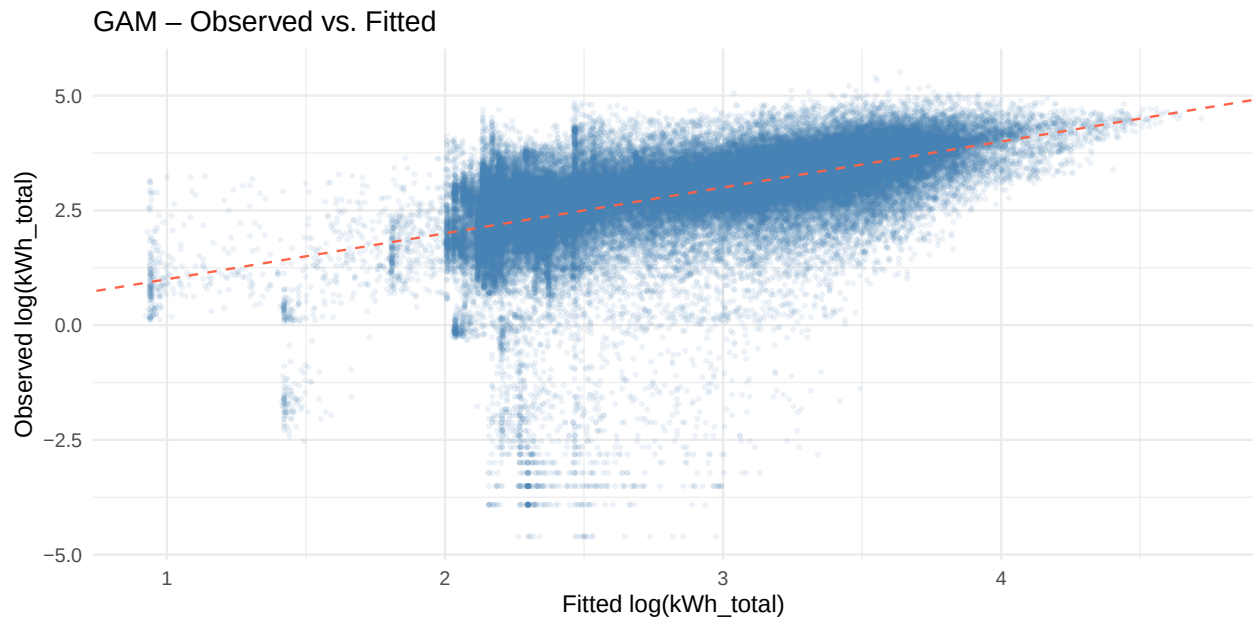
Other predictors held at median / reference level



This plot traces predicted consumption across the full temperature range. A monotone decreasing curve confirms that colder days drive higher consumption through heating demand. A U-shape at the warm end would indicate a summer cooling effect, where electric heat pumps run for cooling rather than heating. The width of the confidence band reveals where the model is most uncertain, which is typically at temperature extremes where fewer observations exist.

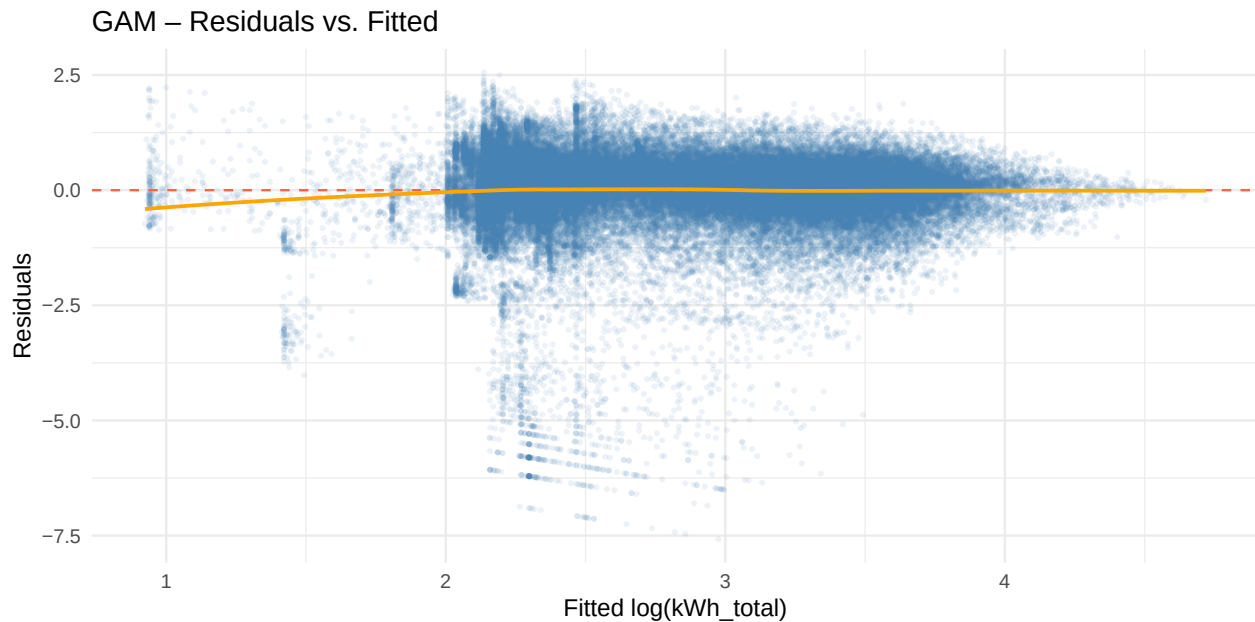
Model Diagnostics

```
fit_df <- data.frame(  
  observed = gam_train$log_kWh_total,  
  fitted   = fitted(model_gam)  
)  
ggplot(fit_df, aes(x = fitted, y = observed)) +  
  geom_point(alpha = 0.1, size = 0.6, color = "steelblue") +  
  geom_abline(intercept = 0, slope = 1, color = "tomato", linetype = "dashed") +  
  labs(title = "GAM - Observed vs. Fitted",  
       x = "Fitted log(kWh_total)", y = "Observed log(kWh_total)") +  
  theme_minimal()
```



The dashed line is the ideal 1:1 line. A perfectly calibrated model would scatter all points along it symmetrically. Systematic deviations above or below indicate that the model consistently over- or under-estimates in certain consumption ranges. A fan shape with wider scatter at high fitted values would suggest that high-consumption days are inherently harder to predict, likely due to household-specific behaviour that the model cannot capture from weather and building characteristics alone.

```
resid_df <- data.frame(
  fitted = fitted(model_gam),
  residuals = residuals(model_gam)
)
ggplot(resid_df, aes(x = fitted, y = residuals)) +
  geom_point(alpha = 0.1, size = 0.6, color = "steelblue") +
  geom_hline(yintercept = 0, color = "tomato", linetype = "dashed") +
  geom_smooth(method = "loess", se = FALSE, color = "orange", linewidth = 0.8) +
  labs(title = "GAM - Residuals vs. Fitted",
       x = "Fitted log(kWh_total)", y = "Residuals") +
  theme_minimal()
```



A well-fitting model shows residuals scattered randomly around zero — the orange smoother (loess) should be approximately flat. A curved smoother would indicate the model still misses a systematic pattern. A funnel shape (residual variance increasing with fitted values) would suggest heteroscedasticity: the model is less precise for high-consumption households, which is common when household-level effects are not explicitly modelled.

```
pred_gam <- predict(model_gam, newdata = gam_test)
actual_log <- gam_test$log_kWh_total
actual_kwh <- exp(actual_log)
pred_kwh <- exp(pred_gam)

mae_kwh <- mean(abs(actual_kwh - pred_kwh))
median_kwh <- median(actual_kwh)
mae_pct <- mae_kwh / median_kwh * 100
r2_gam <- 1 - sum((actual_log - pred_gam)^2) / sum((actual_log - mean(actual_log))^2)

cat(sprintf(
  "R\u00b2 (test):           %.1f%%\nMAE (kWh/day):           %.1f kWh\nMedian consumption: %.1f kWh/day\nMAE as % of median: %.1f%%\nr2_gam * 100, mae_kwh, median_kwh, mae_pct
))
```

```
## R² (test):           30.3%
## MAE (kWh/day):      10.4 kWh
## Median consumption: 20.9 kWh/day
## MAE as % of median: 49.8%
```

The most interpretable performance metric here is the **mean absolute error in kWh**. On average, the model's daily consumption forecasts are off by **10.4 kWh**, against a median household consumption of **20.9 kWh/day**. This means a typical prediction error of about **50%** of a household's usual daily consumption. The model captures broad patterns such as cold days, large houses, and heat pump type, but cannot account for individual household behaviour like occupancy, working from home, or appliance usage on a given day.

The **R² of 30.3%** tells the same story from a different angle. The five predictors explain roughly a third of the day-to-day variation in consumption, and the remaining two-thirds is driven by factors not available

in this dataset. For a client, this means the model is useful for segmentation and benchmarking such as flagging atypically high-consuming households, but should not be used as a precise daily forecast without household-level data.

A note on why RMSE was not used: RMSE on the log scale (0.87) is mathematically correct but has no intuitive unit that a client can relate to. MAE in kWh is directly actionable, and R^2 communicates model quality in plain percentage terms.

Summary: The GAM reveals that temperature and living area have genuinely non-linear effects on electricity consumption, which a plain linear model would misrepresent. Heating degree days, however, turn out to be linear, suggesting the LM’s treatment of that variable was already adequate. Houses consume 11.3% more than apartments, and ground-source heat pumps consume 12.5% less than air-source systems. With an R^2 of 30.3% and a mean prediction error of 10.4 kWh/day, the model performs moderately well for a purely weather- and building-based specification and will serve as the interpretable benchmark in the model comparison section.

Neural Network (NN)

Lead: Raphaël Spagolla

Objective

The aim of this analysis is to predict whether an observation corresponds to a high-consumption event (`high_consumption`), a binary outcome taking values 0 or 1. This is a classification problem, where the focus is on accurately distinguishing between high- and low-consumption cases.

From a practical perspective, reliable classification of high-consumption events can support decision-making processes such as resource allocation, cost management, or demand planning. Early identification of such events allows for more efficient operational responses.

A neural network is employed to capture potentially complex and non-linear relationships between consumption and predictors, including weather conditions, building characteristics, and household composition. Model performance is evaluated based on predictive accuracy and classification metrics derived from test data.

Data Preparation

The dataset consists of 99,838 observations and includes variables related to weather conditions, building characteristics, and household composition. The response variable, *high_consumption*, is binary and approximately balanced between the two classes.

To ensure valid model evaluation, the data is split into training (80%) and test (20%) sets using a stratified sampling approach. This preserves the class distribution in both subsets and avoids biased performance estimates that could arise from imbalanced splits.

Categorical variables are transformed into dummy variables using one-hot encoding, allowing them to be used as inputs in the neural network. In addition, all predictors are scaled based on the training data. This step is important for neural networks, as it improves numerical stability and facilitates efficient training. The same transformations are then applied to the test set to ensure consistency and prevent data leakage.

Model setup

```
cat(deparse(formula(mod_nn)), "\n\n")
```

```
## high_consumption ~ heating_degree_days + temp_avg + living_area + building_typehouse + heatpump
```

```
mod_nn$call
```

```
## nnet.formula(formula = high_consumption ~ ., data = train_nn,  
## size = 3, decay = 0.001, maxit = 300, trace = FALSE)
```

The neural network is specified with an architecture of (8–3–1), consisting of eight input nodes, one hidden layer with three neurons, and a single output neuron for binary classification. This relatively simple structure is chosen to balance model flexibility with the risk of overfitting, allowing the network to capture non-linear relationships without introducing unnecessary complexity.

A non-linear activation function is applied in the hidden layer, enabling the model to learn complex patterns in the data. The output layer produces predicted probabilities for the high-consumption class.

To improve generalisation, weight decay (set to 0.001) is included as a regularisation mechanism. This penalises large weights and helps reduce model variance. The network is trained using the `nnet` package with a sufficient number of iterations to ensure convergence.

Overall, the chosen specification reflects a trade-off between predictive performance and model stability, favouring a parsimonious architecture that performs reliably on unseen data.

Model Results

The neural network achieves an accuracy of approximately 0.77, indicating solid predictive performance and a clear improvement over a naive classifier (≈ 0.50) given the balanced dataset. This suggests that the model captures meaningful structure in the data.

```
# Compute predictions from loaded model  
pred_prob <- predict(mod_nn, newdata = test_nn, type = "raw")  
  
if (is.matrix(pred_prob)) {  
  if ("1" %in% colnames(pred_prob)) {  
    pred_prob_1 <- pred_prob[, "1"]  
  } else {  
    pred_prob_1 <- pred_prob[, ncol(pred_prob)]  
  }  
} else {  
  pred_prob_1 <- pred_prob  
}  
  
pred_class <- ifelse(pred_prob_1 >= 0.5, "1", "0")  
pred_class <- factor(pred_class, levels = levels(test_nn$high_consumption))  
truth      <- test_nn$high_consumption  
  
cm <- table(Predicted = pred_class, Actual = truth)  
acc <- mean(pred_class == truth)
```

```

tp <- cm["1", "1"]; tn <- cm["0", "0"]
fp <- cm["1", "0"]; fn <- cm["0", "1"]

sensitivity <- tp / (tp + fn)
specificity <- tn / (tn + fp)

# ROC object (used by the plots below)
pred_roc <- ROCR::prediction(pred_prob_1, truth)
perf_roc <- ROCR::performance(pred_roc, "tpr", "fpr")

# Display metrics table
results_nn <- data.frame(
  model      = "Neural Network",
  accuracy   = round(acc, 3),
  sensitivity = round(sensitivity, 3),
  specificity = round(specificity, 3)
)

knitr::kable(
  results_nn,
  digits = 3,
  col.names = c("Model", "Accuracy", "Sensitivity", "Specificity"),
  caption = "Neural Network Performance Metrics"
)

```

Table 2: Neural Network Performance Metrics

Model	Accuracy	Sensitivity	Specificity
Neural Network	0.77	0.803	0.737

Sensitivity (0.803) exceeds specificity (0.737), indicating that the model is more effective at identifying high-consumption events than low-consumption cases. This suggests that patterns associated with high consumption are more strongly captured by the available predictors. While this may be advantageous in contexts where detecting high-consumption events is particularly important, it also implies a slightly higher rate of false positives.

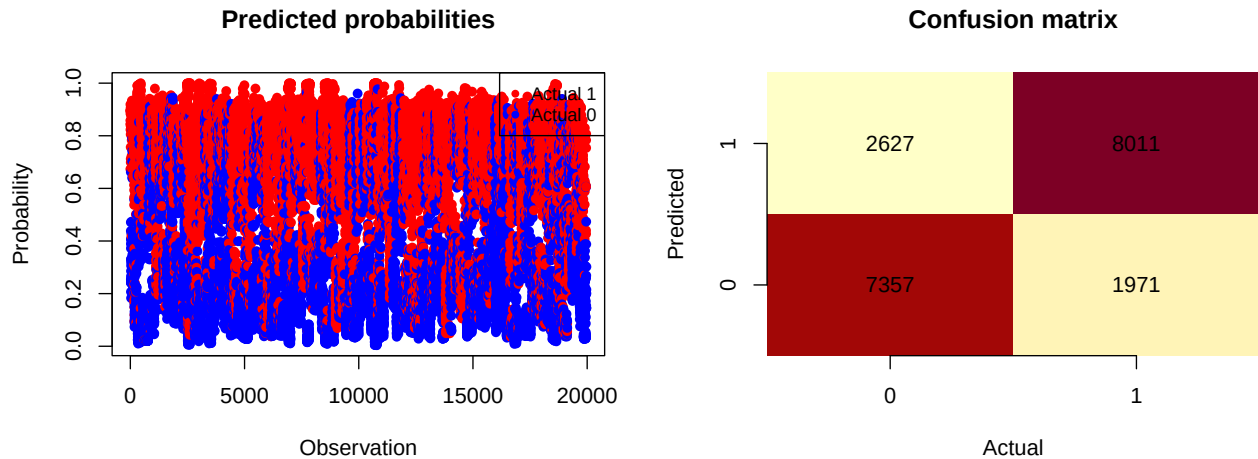
Overall, the confusion matrix shows that the majority of observations are correctly classified, with most predictions concentrated along the diagonal. However, misclassifications occur in both directions, indicating that some observations share similar characteristics across classes.

These results indicate good overall classification performance, although some variability remains unexplained, likely due to overlap in predictor patterns between the two classes.

Visual Evidence

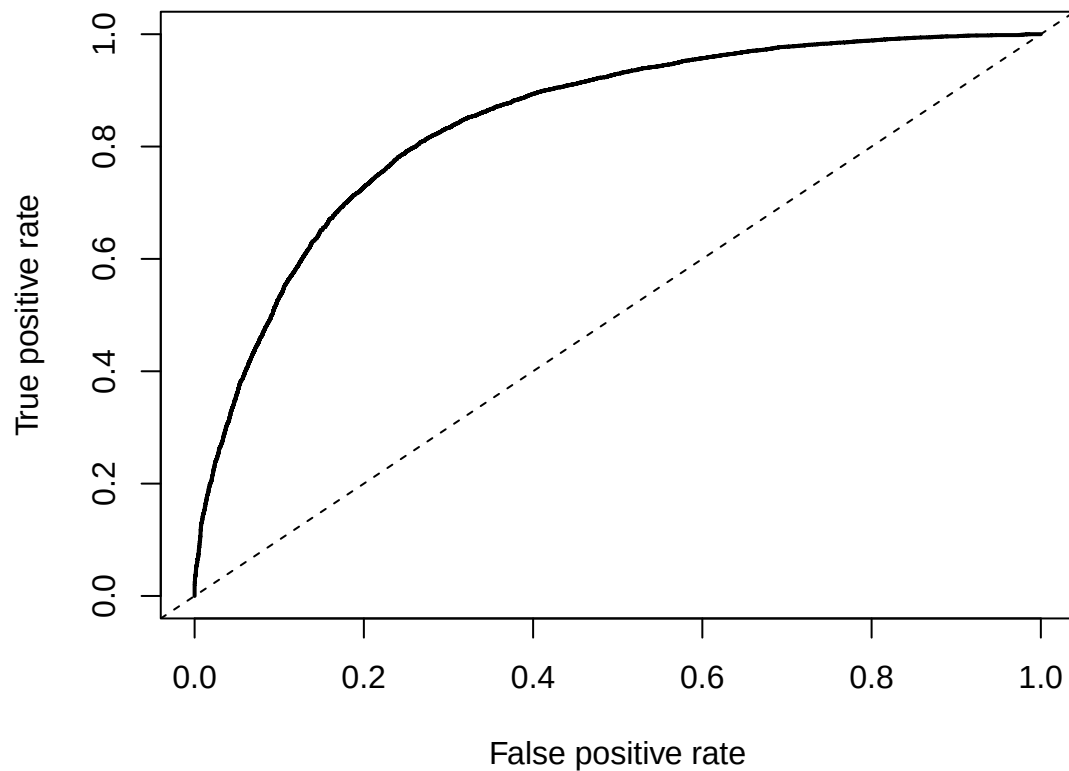
The *predicted probability plot* illustrates how the neural network separates the two classes. Observations belonging to the high-consumption class tend to have higher predicted probabilities, while low-consumption cases are concentrated at lower values. However, noticeable overlap in the mid-range indicates that some observations cannot be clearly distinguished, which explains the presence of classification errors.

The *confusion matrix* provides a direct summary of classification performance. Most observations lie on the diagonal, confirming a high number of correct predictions for both classes. Nevertheless, errors occur in both directions, highlighting that class separation is not complete.



The *ROC curve* further evaluates model performance across all possible classification thresholds. The curve lies clearly above the diagonal reference line, indicating strong discriminative ability and performance well above random classification. The steep initial increase suggests that high true positive rates can be achieved with relatively low false positive rates, while the flattening at higher values reflects diminishing gains.

ROC curve



Model Diagnostics

Neural network diagnostics focus on predictive performance rather than distributional assumptions. The chosen architecture (8–3–1) provides a balance between flexibility and overfitting, allowing the model to capture non-linear relationships while maintaining stable generalisation.

The test accuracy of approximately 77% indicates reasonable generalisation to unseen data. The absence of

a substantial gap between training and test performance suggests that overfitting is limited, although some dependence on the specific train–test split may remain.

The ROC curve confirms consistent performance across classification thresholds and a clear improvement over random classification. However, the observed overlap in predicted probabilities indicates that the available predictors do not fully separate the two classes. This suggests that model performance is constrained more by the information content of the data than by the chosen model structure.

Alternative network configurations were explored to assess the impact of model complexity. Increasing the number of hidden neurons led to slight improvements in training performance but did not translate into meaningful gains in test accuracy, indicating potential overfitting. Simpler architectures achieved comparable results with greater stability, supporting the choice of the 8–3–1 specification.

Key Findings

- The neural network achieves solid predictive performance, with an accuracy of approximately 77%, clearly outperforming a naive baseline.
- The model is more effective at identifying high-consumption events than low-consumption cases, as reflected in higher sensitivity compared to specificity.
- The confusion matrix and ROC curve confirm good overall classification ability, although not perfect.
- Overlap in predicted probabilities indicates that some observations cannot be clearly separated, limiting maximum achievable performance.
- Increasing model complexity does not lead to substantial improvements, suggesting that performance is primarily constrained by the available features rather than the model itself.

Conclusion

The neural network achieves solid predictive performance and successfully captures non-linear relationships in the data. With an accuracy of approximately 77%, it provides a meaningful improvement over baseline classification and demonstrates good overall discriminative ability.

However, the presence of overlap in predicted probabilities indicates that the model cannot fully distinguish between the two classes. This suggests that performance is primarily constrained by the available predictors rather than by model complexity.

From a practical perspective, the model can serve as a useful decision-support tool for identifying high-consumption events, although predictions should be interpreted with some caution due to remaining classification uncertainty.

Further improvements are more likely to come from enhanced feature engineering or the inclusion of additional relevant variables rather than increasing model complexity. In addition, more robust evaluation methods, such as cross-validation, could provide more stable performance estimates and reduce dependence on a single train–test split.

Support Vector Machine (SVM)

Lead: Tharrmeehan Krishnathasan

```
# sum_model <- sum(response ~ predictor1 + predictor2 + ...,  
#                   data = heapo, kernel = "radial", cost = 1, gamma = 0.1)  
# saveRDS(sum_model, "../models/sum_model.rds")  
#sum_model <- readRDS("../models/sum_model.rds")
```

Coefficient Interpretation

Model Comparison

```
# Example: 5-fold CV for all models targeting the same response variable  
# Define CV folds once and reuse for all models
```

```
# Results table: model name | CV-RMSE (or AUC, log-loss, etc.)
```

```
# Bar chart or dot plot comparing CV performance across models
```

Conclusions

Use of Generative AI